



Lesson 3: Motion, Looks, and Sound — Your Sprite's 3 Superpowers

Hi, coder! 🙌

Remember when you made your sprite move and talk for the first time? Today you're leveling up with THREE superpowers working together: **Motion**, **Looks**, and **Sound**.



Think Like a Coder: The Stage Map

The Scratch stage is like a giant map. Every spot has an address made of two numbers: - **x** tells you left or right. Negative numbers go left, positive numbers go right. - **y** tells you up or down. Negative numbers go down, positive numbers go up. - The very middle of the stage is **x: 0, y: 0**.

The "go to x: ___ y: ___" block teleports your sprite straight to an address. The "move ___ steps" block just walks it forward a little at a time, in the direction it's facing.

Your 3 Superpowers

- **Motion blocks** (blue) — your sprite's feet! Move, turn, or teleport.
- **Looks blocks** (purple) — your sprite's face and clothes! Say things, switch costumes, get bigger or smaller.
- **Sound blocks** (pink) — your sprite's voice! Play sound effects and music.

Try This: Build a Sprite Show

1. Add a **"go to x: -100 y: 0"** block — this is your sprite's starting spot.
2. Add a **"move 50 steps"** block underneath.
3. Add a **"next costume"** block.
4. Add a **"play sound [pop] until done"** block.
5. Add a **"say [Wheee!] for 2 seconds"** block.
6. Click the whole stack and watch your sprite's mini show!



Think Like a Coder: Order Matters!

Code runs from TOP to BOTTOM, just like reading a book. Try swapping the order of two blocks in your stack — what changes? This is one of the biggest secrets of coding: **the order you put things in changes what happens.**



Bonus Challenge

Make your sprite look like it's really walking! Try: move a little bit, then switch costume, then move a little more, then switch costume again. Watch it "walk" across the stage.



Debug Detective

Uh-oh! Someone's sprite is supposed to get a little bigger every time it's clicked, but now it's HUGE — way bigger than the stage! What do you think went wrong? (Hint: try adding a "set size to 100" block to reset it back to normal before it grows again.)



Victory Moment

You just gave your sprite three real superpowers at once! Next time, you'll learn how sprites can REACT to things — like key presses and clicks — using events and broadcasting. See you in Lesson 4! 🚀